



VPC Traffic Flow and Security

IR

irahimsindhu@gmail.com

The screenshot displays the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and user information (Europe (Stockholm), Rahim-Abbas-940 (666729139927)). The main content area shows the 'VPC dashboard' on the left sidebar with a 'Filter by VPC' dropdown. The central pane displays the details for the security group 'sg-02e2ce92f733217ec - NextWork Security Group'. A green notification banner at the top of the pane states: 'Security group (sg-02e2ce92f733217ec | NextWork Security Group) was created successfully'. Below this, the 'Details' section shows the security group name, ID, description, owner, and rule counts. The 'Inbound rules' tab is selected, showing a table with one rule: 'sgr-0185095c5b4de6178' for HTTP traffic on port 80. The bottom of the console shows the footer with copyright information and links to CloudShell, Feedback, and Console Mobile App.

Security group (sg-02e2ce92f733217ec | NextWork Security Group) was created successfully

sg-02e2ce92f733217ec - NextWork Security Group

Details

Security group name NextWork Security Group	Security group ID sg-02e2ce92f733217ec	Description A Security Group for the NextWork VPC.	VPC ID vpc-0545d675fb1d89d3c
Owner 666729139927	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules (1)

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0185095c5b4de6178	IPv4	HTTP	TCP	80



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual private network in the cloud that lets you isolate and control your networking environment; it's useful because it provides secure, customizable networking for AWS resources.

How I used Amazon VPC in this project

I used Amazon VPC to create and configure isolated subnets, control traffic with route tables and security groups, and manage how resources communicate with each other and the internet.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was the absolute simplicity and clear guidance.

This project took me...

This project took me 50 minutes to complete.



Route tables

Route tables act like navigation instructions for network traffic. They contain rules that direct packets destined for specific IP address ranges to the appropriate destination, such as a subnet, gateway, or router.

Routes tables are needed to make a subnet public because the route table contains a route (typically 10.0.0.0/16) to an Internet Gateway, which allows the traffic to reach and return to internet.

The screenshot shows the AWS Management Console interface for a specific route table. At the top, a green notification bar indicates that routes were updated successfully. The main heading is 'rtb-06dc000a087c383fa / NextWork Route Table'. Below this, the 'Details' tab is selected, showing information such as the Route table ID, VPC, and Owner ID. To the right, there are sections for 'Main', 'Explicit subnet associations', and 'Edge associations'. Below the details, there are tabs for 'Routes', 'Subnet associations', 'Edge associations', 'Route propagation', and 'Tags'. The 'Routes' tab is active, displaying a table with two routes. The first route has a destination of 0.0.0.0/0 and a target of igw-0f07b89fd7d4a93b1, with a status of 'Active'. The second route has a destination of 10.0.0.0/16 and a target of 'local', also with a status of 'Active'. The console footer shows the year 2026 and various links like CloudShell, Feedback, and Console Mobile App.

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0f07b89fd7d4a93b1	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table



Route destination and target

Destination specifies the IP range where traffic is headed, while Target specifies the next hop (such as a router, gateway, or network interface) that the traffic should be sent to.

The route in my route table that directed internet-bound traffic to my internet gateway had a destination of 0.0.0.0/0 and the target is my public subnet located inside my VPC.

Updated routes for rtb-06dc000a087c383fa / NextWork Route Table successfully

rtb-06dc000a087c383fa / NextWork Route Table

Details

Route table ID: rtb-06dc000a087c383fa

VPC: vpc-0545d675fb1d89d3c | Rahim-NextWor-VPC

Main: Yes

Owner ID: 666729139927

Explicit subnet associations: -

Edge associations: -

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0f07b89fd7d4a93b1	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table



Security groups

Security groups are the rules that monitor the inbound and the outbound traffic that can access our instance and the data that can be sent over to any third-party service from our instance.

Inbound vs Outbound rules

Inbound rules define what incoming traffic is allowed to reach our instance(such as a server or any other instance). The inbound rule of my security group are: - Protocol: TCP - Type: HTTP - Port: 80 - Source: 10.0.0.0/0 - IP Version: Ipv4

Outbound rules define what outgoing traffic a resource is allowed to send to the other destinations like a server, third-party server or an instance. Outbound rules of my security group are: - Protocol: All - Type: All traffic - Port: All - Source: 0.0.0.0/0 - IP Version: Ipv4



VPC dashboard

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Security group (sg-02e2ce92f733217ec | NextWork Security Group) was created successfully

Details

sg-02e2ce92f733217ec - NextWork Security Group

Actions

Details

Security group name

NextWork Security Group

Security group ID

sg-02e2ce92f733217ec

Description

A Security Group for the NextWork VPC.

VPC ID

vpc-0545d675fb1d89d3c

Owner

666729139927

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Search

	Name	Security group rule ID	IP version	Type	Protocol	Port range
☐	-	sgr-0185095c5b4de6178	IPv4	HTTP	TCP	80

CloudShell

Feedback

Console Mobile App

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Network ACLs

Network ACLs are used to set broad traffic rules that apply to an entire subnet. For example, blocking incoming traffic from a particular range of IP addresses or denying all outbound traffic to certain ports.

Security groups vs. network ACLs

Security Groups are attached to individual resources (such as EC2 instances), whereas Network ACLs are attached to subnets and apply to all resources within those subnets.



Default vs Custom Network ACLs

Similar to security groups, network ACLs use inbound and outbound rules

By default, a network ACL's inbound and outbound rules will allow all traffic whether it be inbound or outbound. The details of the inbound and outbound rules is: - Rule Number: 100 - Type: All Traffic - Protocol: All - Source: 0.0.0.0/0 - Allow/Deny: Allow

In contrast, a custom ACL's inbound and outbound rules are automatically set to deny all the traffic whether it be outbound or inbound. The details of the inbound and outbound rules is: - Rule Number: * - Type: All Traffic - Protocol: All - Source: 0.0.0.0/0 - Allow/Deny: Deny

Network ACLs (1/3)

Name	Network ACL ID	Associated with	Default	VPC ID
-	acl-0f4a4af223f519f41	subnet-00e791d1ad4dc15c	Yes	vpc-0e0f665501ef04fe0
-	acl-03c392a4d527bbcbd	-	Yes	vpc-0545d675fb1d89d3c / Ra...
NextWork Network A...	acl-0920518ba3d621878	subnet-05b5b416ee77a2010 / NextWork Subnet	No	vpc-0545d675fb1d89d3c / Ra...

acl-0920518ba3d621878 / NextWork Network ACL

Inbound rules (2)

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny



Tracking VPC Resources

I created additional VPC, Internet Gateways. Instead of my usual region, I used "eu-west-3". Teams would use multiple regions: - To improve availability, - Reduce latency for users in different regions, - Ensure disaster recovery if one server goes down.

EC2 Global View is a tool where you can find all the available Amazon Web Services EC2 resources across regions. I could even narrow down my search by region, instance state, tags, or instance type. Without EC2 Global View, you'd have to switch between regions one by one manually to find and compare resources.

You would use Amazon Web Services EC2 Global View when you need a single, centralized view of EC2 resources across multiple regions, especially to find, compare, or audit instances quickly without switching incs regions manually.

AWS Global View

Region Explorer

Global search

Regions and Zones

Settings

Region explorer (33)

The region explorer lists your resources across all Regions for which your account is enabled.

Updated 1 minute ago

[Open selected Region](#) [View Region details](#) [View Region resources](#)

Filter region using region name

Region	Instances	VPCs	Subnets	Security Groups	Volumes	Aut
☐ Asia Pacific (Singapore) ap-southeast-1	–	1	3	1	–	–
☐ Asia Pacific (Sydney) ap-southeast-2	–	1	3	1	–	–
☐ Asia Pacific (Tokyo) ap-northeast-1	–	1	3	1	–	–
☐ Canada (Central) ca-central-1	–	1	3	1	–	–
☐ Europe (Frankfurt) eu-central-1	–	1	3	1	–	–
☐ Europe (Ireland) eu-west-1	–	1	3	1	–	–
☐ Europe (London) eu-west-2	–	1	3	1	–	–
☐ Europe (Paris) eu-west-3	–	2	3	3	–	–
☐ Europe (Stockholm) eu-north-1	–	2	2	3	–	–
☐ South America (São Paulo) sa-east-1	–	1	3	1	–	–
☐ United States (N. California) us-west-1	–	1	2	1	–	–
☐ United States (N. Virginia) us-east-1	–	1	6	1	–	–

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